Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/15/2007
PI Name:	Ladd, Anthony J.C. Ph.D.		
Project Title:	Dynamics and Structures in Colloidal Suspensions far from Equilibrium		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI has completed submission (kw 11/30)		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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We will carry out a coordinated sequence of numerical simulations and laboratory experiments to address fundamental questions related to the dynamics of batch sedimentation, fluidized beds, and centrifugal flows. In some instances the crucial scientific issues can be addressed by ground-based experiments, supported by simulations. However, there are other questions that can only be answered in a microgravity environment. We propose to simulate suspension flows in the absence of gravity to obtain partial answers to these questions in the near term, and also to guide a future flight experiment. The proposed work follows on from current NASA-supported investigations, which will end in November 2003.

The objective of this work is to continue our ongoing efforts to develop a better understanding of gas fluidized systems by extending the current limits of laboratory experiments and numerical simulation. We use experiments and

Task Description:	simulations synergistically; thus we not only use laboratory experiments to validate the results of numerical simulations, but we also use simulations to clarify aspects of laboratory experiments, such as polydispersity, that are difficult to control experimentally. Numerical simulations will also be used to provide preliminary answers to questions that can only be fully investigated in a microgravity environment. Specifically we will attempt to answer three fundamental questions related to particle-fluid flows, motivated in part by what was discovered in the past 3 years research. The first question is to understand the origin of the observed correlation length in settling suspensions and decide if it is universal, applying also to fluidized suspensions. The second issue is to discover if particle collisions in a low-Reynolds number fluidized bed lead to important changes in suspension microstructure at finite Stokes numbers, and to investigate whether such microstructural changes are related to the onset of instability in gas-fluidized suspensions at low Stokes numbers. Numerical simulations will be used to uncover the fundamental physics underlying the effects of small amounts of particle inertia, which under normal conditions is masked by the effects of the earth's gravitational field. Although it is possible to study liquid-fluidized suspensions at finite-Stokes numbers in the laboratory, the parameter ranges that characterize gas-fluidized flows can only be achieved in a reduced gravity environment. Finally we will examine a flow with a non-uniform external force, namely a rotating suspension flow. In this case the centrifugal force produces a non-uniform body force on the particles, leading to a surprising segregation of particles along the rotation axis into bands of high and low concentration. In this case it is essential to remove gravity altogether, so that the effects of the centrifugal force can be studied in isolation. Moreover, the bands of high and low concentration may develop differently in a microgravity environment than on earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Inertial migration of neutrally buoyant particles in a square duct has been investigated by numerical simulation in the range of Reynolds numbers from 100 to 1000. Particles migrate to one of a small number of equilibrium positions in the cross-sectional plane, located near a corner or at the center of an edge. In dilute suspensions, trains of particles are formed along the axis of the flow, near the planar equilibrium positions of single particles. At high Reynolds numbers (Re > 750), we observe particles in an inner region near the center of the duct. We present numerical evidence that closely-spaced pairs of particles can migrate to the center at high Reynolds number. Several different interpolation schemes have been proposed for improving the accuracy of lattice-Boltzmann simulations in the vicinity of a solid boundary. However these methods require at least two or three fluid nodes between nearby solid surfaces, a condition that may not be fulfilled in dense suspensions or porous media for example. Here we propose an interpolation of the equilibrium distribution, which leads to a velocity field that is both second-order accurate in space and independent of viscosity. The Equilibrium Interpolation Rule infers population densities on the boundary itself to reduce the span of nodes needed for interpolation; it requires a minimum of one grid spacing between the nodes. By contrast, the Linear Interpolation Rule requires two fluid nodes in the gap and leads to a viscosity-dependent slip velocity, while the Multi-Reflection Rule is viscosity independent but requires a minimum of three fluid nodes.
Bibliography Type:	Description: (Last Updated: 03/15/2007)
Articles in Peer-reviewed Journals	Chun B, Ladd AJC. "Inertial migration of neutrally buoyant particles in a square duct: An investigation of multiple equilibrium positions." Phys. Fluids 2006 Mar;18(3):031704. http://dx.doi.org/10.1063/1.2176587 , Mar-2006